1.11 Quiz: Statistics — Markscheme

1. [Maximum mark: 7]

   Mid-interval values $5, 15, 25, 35, 45$ with frequencies $4, 12, 18, 11, 5$; $n = 50$.

   GDC: *enter the mid-interval values in one list and the frequencies in a second list, then run* `One-Variable Statistics` *with the frequency list.*

   (a) Continuous **A1**

   Time is measured, not counted. Accept a reason such as "time can take any value in an interval"; no reason is required.

   (b) $20 \leq t < 30$ **A1**

   *The class, not its frequency: "*18*" alone earns A0. Accept "*20 *to* 30 *minutes". The mid-interval value* 25 *alone earns A0.*

   (c) $\bar{x} = \dfrac{4(5) + 12(15) + 18(25) + 11(35) + 5(45)}{50}$ **(M1)**

   $= \dfrac{1260}{50} = 25.2$ minutes **A1**

   M1 for using the mid-interval values with the frequencies, shown as a sum or as the GDC lists. Using the lower class boundaries ($0, 10, 20, 30, 40$*, giving* 20.2*) or upper boundaries (giving* 30.2*) earns M0A0. The mean of the frequencies,* 10*, earns no marks.*

   (d) $\sigma = 10.860\ldots = 10.9$ minutes (3 s.f.) **A1**

   Do not accept $s_x = 10.971\ldots$ *(*11.0*), the sample standard deviation.*

   (e) $\dfrac{192}{600} \times 50$ **M1**

   $= 16$ patients **A1**

   M1 for the over-65 *proportion of the registered patients applied to* 50*, however written (e.g.* $\frac{50}{600} = \frac{1}{12}$*, then* $\frac{192}{12}$*).*

2. [Maximum mark: 9]

Ordered: $8, 22, 24, 25, 26, 27, 28, \mathbf{28}, 29, 30, 31, 32, 33, 35, 36 \quad (n = 15)$.

GDC: *the TI-Nspire excludes the median from each half, so Q_1 is the median of the lowest seven values and Q_3 the median of the highest seven.*

(a) Median $= 28$ **A1**

$Q_1 = 25, \quad Q_3 = 32$ **A1**

Both quartiles are needed for the second A1. Accept $Q_1 = 25.5$ and $Q_3 = 31.5$ from a hand method that includes the median in each half; feedback: the GDC convention is the one to use.

(b) IQR $= 32 - 25 = 7$ **A1**

FT: award A1 for $Q_3 - Q_1$ from the candidate's own quartiles. The range, $36 - 8 = 28$, earns no mark.

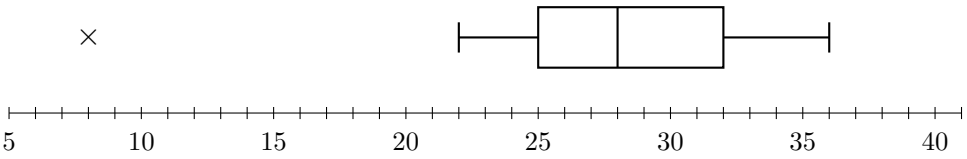
(c) $Q_1 - 1.5 \times \text{IQR} = 25 - 1.5 \times 7$ **M1**

$= 14.5$ **A1**

$8 < 14.5$, so 8 is an outlier. **R1**

M1 for $1.5\times$ their IQR subtracted from their Q_1. R1 for the comparison with 8 stated, not just the fence value. Subtracting $1.5 \times$ IQR from the median or the mean earns no marks. The upper fence, $32 + 10.5 = 42.5$, is not required. FT: award all three marks from the candidate's own Q_1 and IQR, provided their fence is above 8.

(d)



Box from $Q_1 = 25$ to $Q_3 = 32$ with the median line at 28 **A1**

Whiskers to 22 and 36 **A1**

Outlier 8 marked with a cross **A1**

Award each mark independently; plotted values within half a small division (± 0.5) are accepted. A whisker drawn to 8 or to 14.5 loses the second A1. FT: award the first A1 for a box drawn at the candidate's own quartiles from (a).

3. [Maximum mark: 9]

200 fish; the curve passes through $(20, 30)$, $(24, 50)$, $(28, 75)$, $(32, 100)$, $(40, 150)$ and $(44, 170)$.

Readings within one small square of the values below are accepted (± 2 cm on the length axis, ± 5 fish on the cumulative frequency axis), and any value following from the candidate's own readings earns the A1. Feedback: draw the reading lines on the graph.

(a) Median: cumulative frequency 100 gives $L = 32$ cm **A1**

(b) $Q_1 = 24$ (at 50), $\quad Q_3 = 40$ (at 150) **(M1)**

$\text{IQR} = 40 - 24 = 16$ cm **A1**

M1 for reading the lengths at cumulative frequencies 50 *and* 150 *(or for either quartile correct). Reading at* 25 *and* 75 *(percent of* 100*) earns M0.*

(c) $200 - 170$ **(M1)**

$= 30$ fish **A1**

M1 for reading the cumulative frequency at $L = 44$ and subtracting from 200*. The reading* 170 *alone, as a final answer, earns M1A0.*

(d) 15% of $200 = 30$ **(M1)**

Cumulative frequency 30 gives $L = 20$ cm **A1**

M1 for 30 *(or for reading the length at cumulative frequency* 30*). Reading at cumulative frequency* 170 *(the longest* 15%*, $L = 44$) earns M1A0. Reading at cumulative frequency* 15 *(treating the total as* 100*, $L = 16$) earns M0A0.*

(e) $p = 75 - 30 = 45$ **A1**

$q = 170 - 75 = 95$ **A1**

The two marks are independent. Accept $q = 200 - 30 - 45 - 30 = 95$. FT: award the A1 for q from the candidate's own p when q is found by subtraction from 200*.*

4. [Maximum mark: 8]

$n = 8, \quad \bar{x} = 5.5, \quad \bar{y} = 18.875.$

GDC: *enter x and y in two lists and run* `Linear Regression (mx+b)`.

(a) $r = -0.97297\ldots = -0.973$ (3 s.f.) **A1**

The negative sign is required. Do not accept $r^2 = 0.947$, which sits beside r on the regression screen.

(b) Strong, negative (linear) correlation **A1A1**

A1 for "negative", A1 for "strong" (accept "very strong"); the two marks are independent. "Moderate" or "weak" earns A0 for the strength. FT from the candidate's r in (a).

(c) $y = -1.84x + 29.0$ **A1A1**

A1 for $a = -1.84$, A1 for $b = 29.0$; the two marks are independent. Accept $a = -1.8397\ldots$, $b = 28.993\ldots$ (or 29), or either value to more figures. The reversed line $x = -0.515y + 15.2$ earns A0A0.

(d) For each additional year of age, the price decreases by about 1.84 thousand dollars (about \$1840). **A1**

The answer must give the rate (about 1.8 thousand dollars) per year and the direction (decrease), in context. "Older cars are cheaper" restates the correlation without the rate and earns A0. FT from the candidate's a in (c).

(e) $y = -1.84(7) + 29.0$ **(M1)**

$= 16.115\ldots \approx 16.1$ thousand dollars **A1**

Accept any value consistent with the candidate's own correctly-rounded coefficients, including 16.115… (full precision) and 16.12 (from −1.84 and 29.0), each rounding to 16.1; accept \$16 100. FT from the candidate's line in (c). ***Feedback:*** *store the coefficients in the GDC (or use the regression function* `f1(7)`*) and round only the final answer.*